BENZYLPENICILLIN

Penicillin allergy

1.2 g = 2 million units

CHEMICAL NAME (2S,5R,6R)-3,3-dimethyl-7-oxo-6-[(2-phenylacetyl)amino]-4-thia-1-azabicyclo[3.2.0]heptane-2-carboxylic acid (sodium or potassium salt)

CLASS Natural penicillin (penicillin G), β-lactam antibiotic

TRADE NAMES Crystapen · Penicillin G sodium · Pfizerpen

THERAPEUTIC USES / INDICATIONS

- Streptococcal infections: cellulitis, pneumonia (pneumococcal), endocarditis.
- Meningococcal disease (give before hospital transfer if suspected).
- Syphilis (neurosyphilis), tetanus, gas gangrene, diphtheria, anthrax.
- Group B streptococcal prophylaxis in labour.

ADMINISTRATION

- **IV** slow injection over 3–5 min or infusion; **IM**.
- Not oral (destroyed by gastric acid).
- Rapid injection of large doses can cause seizures.
- Potassium salt: monitor K^+ with high doses.

DOSAGES

- **Adult:** 600 mg–1.2 g IV every 6 h; severe up to 2.4 g every 4 h (14.4 g/day).
- **Meningococcal (pre-hospital):** adult/child >10 yr 1.2 g IV/IM; child 1–9 yr 600 mg; <1 yr 300 mg.
- **Children 1 month–12 yr:** 100 mg/kg/day in 4 divided doses (max 4 g/day); meningitis 180–300 mg/kg/day in 4–6 doses.
- **Conversion:** 600 mg = 1 million units.

PHARMACOKINETICS

- **Onset:** IV immediate. **Duration:** short — frequent dosing needed.
- Penetrates inflamed meninges.
- Excreted rapidly in urine (tubular secretion, blocked by probenecid). **t½:** 30 min.

NURSING MANAGEMENT

- Confirm allergy status; observe 30 min after first dose.
- Check dose units (mg vs units) carefully.
- Monitor temperature, cultures, renal function and electrolytes.
- Syphilis: warn about Jarisch–Herxheimer reaction.
- Suspected meningococcal disease: do not delay first dose.

PHARMACODYNAMICS

- Inhibits **cell-wall synthesis** → bactericidal.
- Narrow spectrum: streptococci, meningococci, *Treponema*, *Clostridium*, *Actinomyces*.
- Destroyed by β-lactamase (most *Staph. aureus* resistant).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea.
- Rash, urticaria.
- Injection-site pain.

ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis (higher risk with parenteral penicillin).
- Seizures (high doses, renal failure, intrathecal).
- Jarisch–Herxheimer reaction (syphilis): fever, chills within hours of first dose.
- Electrolyte disturbance (Na^+/K^+ load).

CONTRAINDICATIONS & CAUTIONS

- Penicillin hypersensitivity.
- Never give intrathecally in standard doses.
- **Caution:** renal impairment, epilepsy.

KEY INTERACTIONS

- **Probenecid:** ↑ penicillin levels.
- **Methotrexate:** ↑ toxicity.
- **Tetracyclines:** may antagonise effect.

PATIENT EDUCATION

- Report rash, swelling or wheeze immediately.
- Fever and chills after the first syphilis dose usually settle within 24 h.
- Complete treatment.

KEY POINTS

- Penicillin G; narrow-spectrum, IV/IM only.
- 600 mg = 1 million units.
- Drug of choice: streptococci, meningococci, syphilis.
- Give early in suspected meningococcal sepsis.

SOURCES CHECKED **[113]** Benzylpenicillin sodium 1200mg Powder for Injection (UK SmPC) **[114]** Penicillin G Potassium, IM, IV (US label)

Full references p. 432–457